

DATABASE SYSTEMS [Revised Credit System] (Effective from the academic year 2025-26) SEMESTER - IV			
Subject Code	CSS 2201	IA Marks	50
Number of Lecture Hours/Week	04	Exam Marks	50
Total Number of Lecture Hours	48	Exam Hours	03
CREDITS - 04			
Course objectives: This course will enable students to • To understand the fundamental concepts of relational model • To develop and query the database using Structured Query Language • To understand the Entity Relationship Model for database design • To use the normalization forms in building an effective database schema • To interpret the different indexing and hashing methods and understand transaction and concurrency control concepts			
Module -1			Teaching Hours
INTRODUCTION: Database-System Applications, Purpose of Database Systems, View of Data, Database Languages, Relational Databases, Database Design, Data Storage and Querying, Transaction Management, Database Architecture, Database Users and Administrators. RELATIONAL MODEL: Structure of Relational Databases, Database Schemas, Keys, Schema Diagrams, Relational Query Languages, Relational Operations, Formal Definition of Relational Algebra, Relational Algebra – Fundamental Operations. QUERY PROCESSING: Overview of Query Processing, Measure of Query Cost, Selection Operation, Sorting, Join Operation, Other Operations. Text Book 1: Chapter 1: 1.1- 1.8, Chapter 2: 2.1 – 2.6, Chapter 15: 15.1 -15.6.			8 Hours
Module -2			

STRUCTURED QUERY LANGUAGE: SQL Data Definition, SQL Data Types and Schemas, Integrity Constraints, Basic Structure of SQL Queries, Set Operations, Aggregate Functions, Nested Subqueries, Additional Basic Operations, Null Values, Modification of the Database. Join Expressions, Views, Transactions. Text Book 1: Chapter 3: 3.2 – 3.9, Chapter 4: 4.1 – 4.6.	12 Hours
--	-----------------

Module – 3	
DATABASE DESIGN USING E-R MODEL: Overview of the Design Process, The Entity-Relationship Model, Constraints, Removing Redundant Attributes in Entity Sets, Entity- Relationship Diagrams, Entity-Relationship Design Issues, Extended E-R Features, Reduction to Relational Schemas. NORMALIZATION: Features of Good Relational Design, Atomic Domains and First Normal Form, Decomposition Using Functional Dependencies, Functional Dependency Theory, Algorithms for Decomposition, Decomposition Using Multivalued Dependencies. Text Book 1: Chapter 6: 6.1-6.10.1, Chapter 7: 7.1-7.7	14Hours
Module-4	
INDEXING AND HASHING: File Organization, Organization of Records in Files, Basic concepts, Ordered Indices, B+ Tree Index Files, B+ Tree Extensions, Multiple-Key Access, Static Hashing, Dynamic Hashing, Comparison of Ordered Indexing and Hashing., Bitmap Indices Text Book 1 : Chapter 13: 13.2-13.3, Chapter 14: 14.1 -14.9 TRANSACTION MANAGEMENT: Transaction Concept, A simple Transaction model, Transaction Atomicity and Durability, Transaction Isolation, Serializability, Transaction Isolation and Atomicity, Transaction Isolation Levels, Failure Classification, Storage, Recovery and Atomicity, Recovery algorithm. Text Book 1: Chapter 17: 17.1-17.8, Chapter 19: 19.1- 19.4 CONCURRENCY CONTROL: Lock-based protocols, deadlock handling, Timestamp based protocols, Validation based protocols. Chapter 18:18.1-18.2, 18.5-18.6.	14 Hours
Course outcomes:	

After studying this course, students will be able to:

1. Articulate the usage of database, relational model and organization of files for query based systems.
2. Write and optimize structured query languages for efficient data retrieval, manipulation and management.
3. Design databases using E-R models for a real world scenario.
4. Apply the normalization technique to decompose given relation schema into effective schemas.
5. Interpret the various transaction management, recovery and concurrency control techniques.
6. Explore the latest database systems.

Reference Books:

1. Silberschatz, Korth, Sudarshan, *Database System Concepts, (7e)*, McGrawHill, New York, 2011.
2. Pramod J Sadalage, Martin Fowler, *NoSQL Distilled*, Addison-Wesley, 2013.
3. Ramez Elmasri and Shamkant Navathe, Durvasula V L N Somayajulu, Shyam K Gupta, *Fundamentals of Database Systems, (6e)*, Pearson Education, United States of America, 2011.
4. Thomas Connolly, Carolyn Begg, *Database Systems – A Practical Approach to Design, Implementation and Management, (4e)*, Pearson Education, England, 2005.
5. Peter Rob, Carlos Coronel, *Database Systems–Design, Implementation and Management, (10e)*, Course Technology, Boston, 2013.
6. https://onlinecourses.nptel.ac.in/noc25_cs40/preview Introduction to Database Systems